

# Seed robustness of the gate-unfreeze QFT

100 random-initialisation seeds  $\times$  three unfreeze orderings, top-20% training objective, DIV2K-8q

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## Question

Does the gate-unfreeze QFT endpoint depend on the random draw? We retrain the `QFTBasis(8,8)` operator from **Haar-random** init under the block-growth (**bg**), left $\rightarrow$ right (**lr**) and right $\rightarrow$ left (**rl**) unfreeze orderings, for 100 seeds each. Every seed  $s$  reseeds **everything trainable** — the Haar gate init, the 50-image training-batch subsample (from the fixed 500-image pool), and the training RNG — while the held-out **test set is held fixed** (canonical seed 42) so the endpoint-PSNR spread reflects the trained operator, not the test draw. Training minimises the top-20%-coefficient MSE (the headline elsewhere is top-10%); test PSNR is scored at four keep ratios  $\rho$ .

## The random initialisations are genuinely different

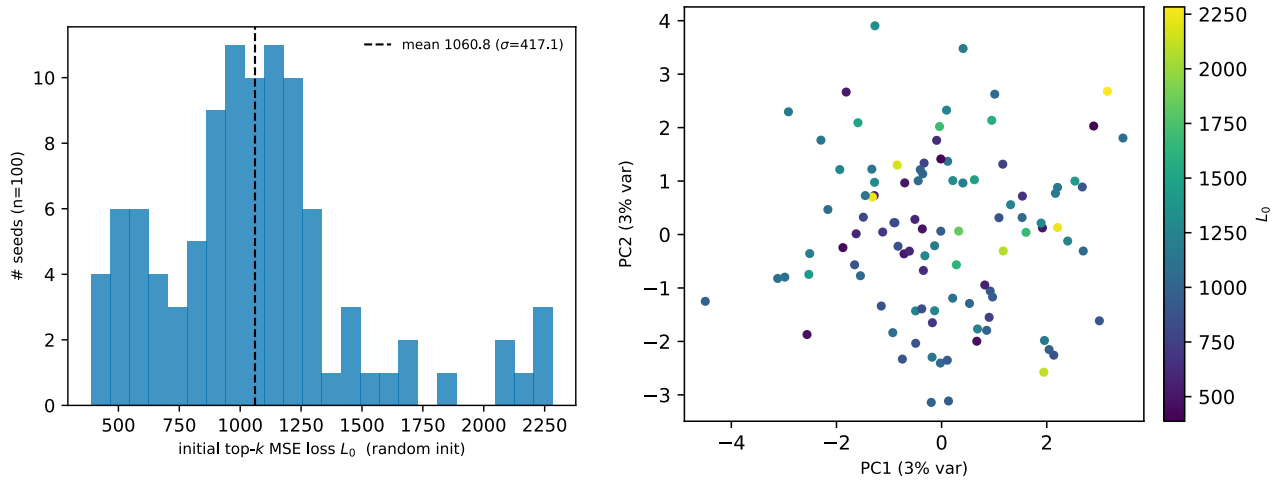


Figure 1: **(a)** the untrained random operator’s initial top- $k$  MSE loss  $L_0$  on a common fixed batch, per seed — it spans 387.3–2283.1 (mean 1060.8,  $\sigma = 417.1$ ), so the seeds start from genuinely different points. **(b)** a 2-D PCA of the 100 init parameter vectors; the top two components explain only 2.8% + 2.6% of the variance, i.e. the inits are spread across many near-orthogonal directions rather than clustered.

## Endpoint variance

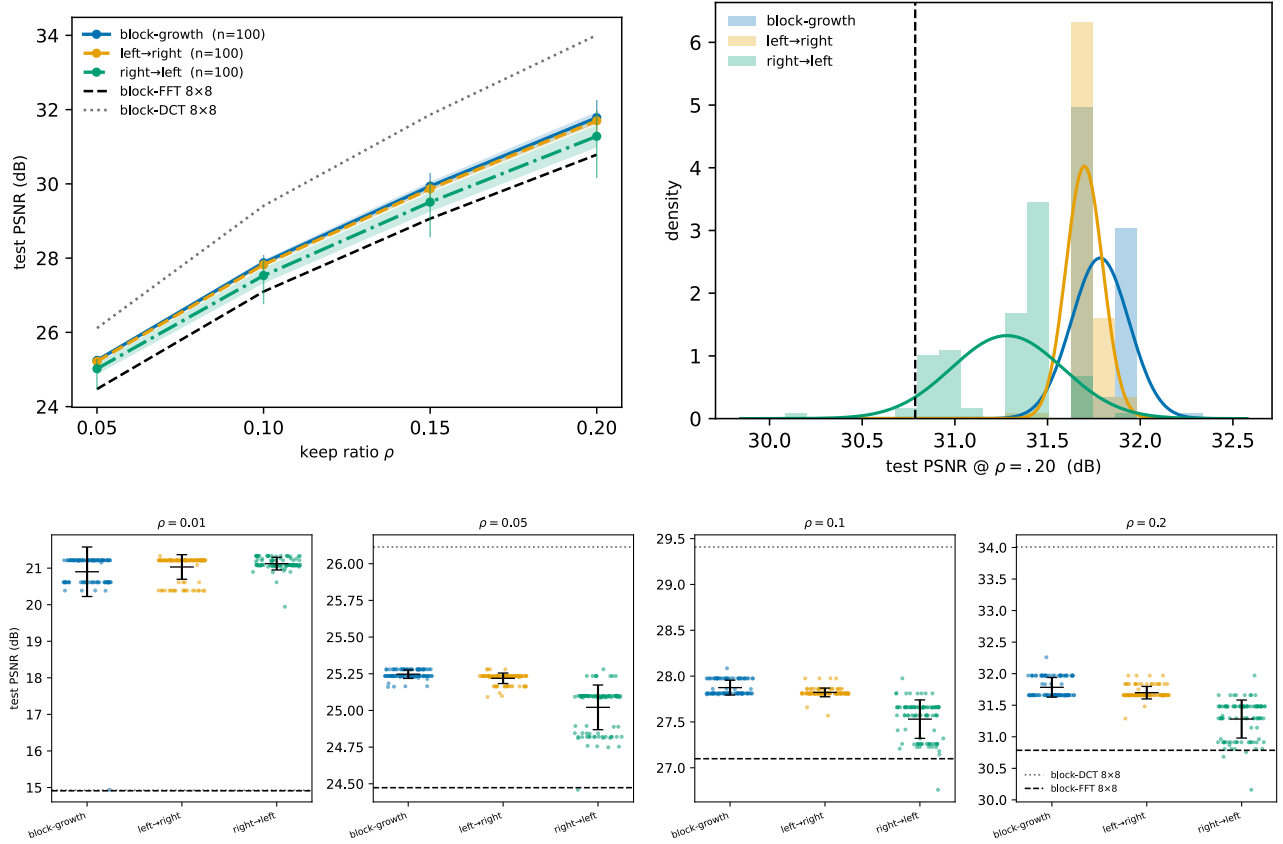


Figure 2: Per-ordering test PSNR across 100 seeds. **Top:** (a) mean  $\pm\sigma$  band (+ min–max whiskers) vs  $\rho$ , against block-FFT 8×8 (dashed) and block-DCT 8×8 (dotted); and (c) the endpoint distribution at  $\rho=\{.20$  (histogram + fitted normal). **Bottom:** (b) per-seed scatter at four keep ratios ( $\rho = 0.01, 0.05, 0.10, 0.20$ ) — the trained bases beat block-FFT at every rate and exceed both classical references by  $\sim 6$  dB at  $\rho=\{.01$ , while block-DCT overtakes them for  $\rho \geq .05$ .

| ordering      | $\rho=\{.05$      | $\rho=\{.10$      | $\rho=\{.15$      | $\rho=\{.20$      | min@.20 |
|---------------|-------------------|-------------------|-------------------|-------------------|---------|
| block-growth  | <b>25.25±0.03</b> | <b>27.87±0.08</b> | <b>29.94±0.12</b> | <b>31.78±0.16</b> | 31.65   |
| left→right    | 25.22±0.04        | 27.82±0.05        | 29.87±0.07        | 31.7±0.1          | 31.29   |
| right→left    | 25.02±0.15        | 27.53±0.21        | 29.51±0.26        | 31.28±0.3         | 30.16   |
| block-FFT 8×8 | 24.47             | 27.1              | 29.06             | 30.79             | —       |
| block-DCT 8×8 | 26.11             | 29.41             | 31.86             | 34.01             | —       |

mean  $\pm\sigma$  test PSNR (dB),  $n = 100$  random-init seeds per ordering; **min@.20** is the single worst seed at  $\rho=\{.20$ .

## Reading

Despite starting from 100 genuinely different random inits (previous section), each ordering’s endpoint sits in a narrow band: at  $\rho=\{.20$  the per-seed standard deviation is block-growth 0.16, left→right 0.1, right→left 0.3 dB. The **release order**, not the random draw, is the dominant factor: block-growth and left→right are tight and sit highest, while right→left is  $\sim 2\times$  wider and lowest — the dynamics figure below explains why (rl starts from a far worse loss and its seeds stay spread until the final unfreeze stages).

Against the classical block-FFT 8×8 reference (30.79 dB @ $\rho=\{.20$ ): block-growth and left→right clear it for **every** seed (worst seed 31.65 / 31.29 dB; margins +0.86 / +0.5 dB). right→left clears it for 97 of 100 seeds; 3 low-tail seeds dip below by at most 0.63 dB. Overall 297 of 300 runs (99%) clear block-FFT. So the gate-unfreeze QFT is **essentially always** above block-FFT — unconditionally for bg/lr, and for 97% of the highest-variance

rl ordering — while block-DCT  $8 \times 8$  (34.01 dB) remains the strongest classical transform here, ahead of the QFT family.

The per-seed endpoints are **not** Gaussian: a Shapiro–Wilk test rejects normality for all three orderings ( $p < 10^{-7}$ ), because seeds settle into a few discrete attractor basins of the very-flat top- $k$  MSE valley rather than scattering smoothly. The fitted normal in the endpoint panel is a visual summary of location and spread, not a claim of Gaussianity.

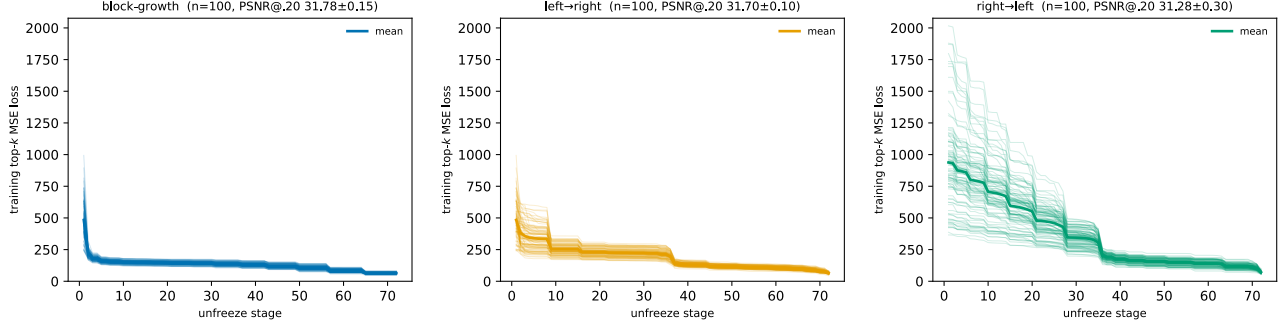


Figure 3: Per-seed training dynamics: every seed’s top- $k$  MSE staircase (one per unfreeze stage), one panel per ordering, on a shared y-range. Different Haar starts descend along different paths: block-growth (a) collapses fast and stays tight, whereas right→left (c) starts far higher and its seeds remain spread until the final stages — visually accounting for the endpoint variances above. left→right (b) is intermediate.

## Emergent block structure of the trained QFT

We close by reading the **canonical published** trained QFT (QFTBasis(8,8), seed 98) directly, making the paper’s central structural claim concrete (sec5.3): the trained full-image QFT factors itself into a block code with no block prior. Of its 16 Hadamard-role gates (8 per dimension), 4 per dimension collapse to non-mixing **Pauli-Z/X** gates (the off-diagonal/diagonal modulus drops to  $\approx 0.3\%$ ) — classical block indices — while 53 of the 56 controlled-phase gates stay active, so the intra-block transform stays rich. The 4 still-mixing row qubits and 4 column qubits leave an effective  $16 \times 16$ -pixel block.

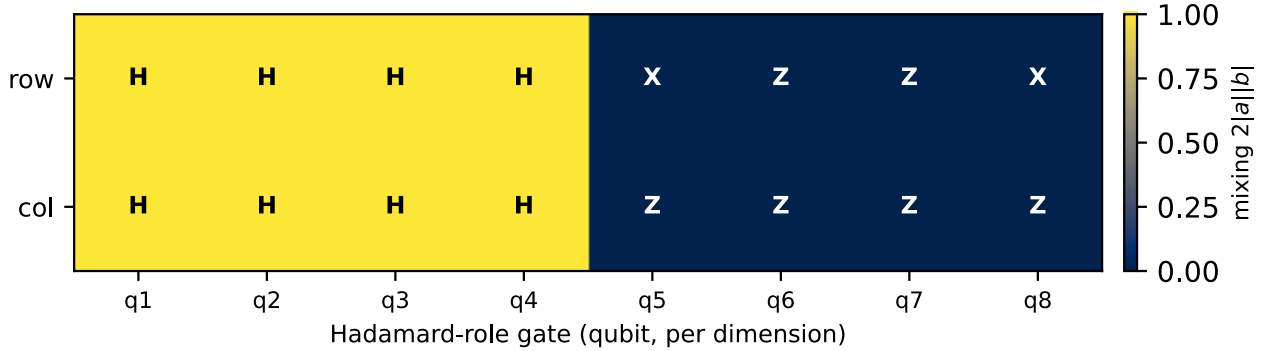


Figure 4: Hadamard-role gates of the trained QFT, one cell per qubit and dimension, shaded by the mixing score  $2|a||b|$  (1 = Hadamard, 0 = frozen) and labelled by type. Four of the eight Hadamards per dimension keep mixing frequencies; the other four freeze to Pauli-Z (or X) and act as classical block indices.

The factorization shows directly in the dense operator.

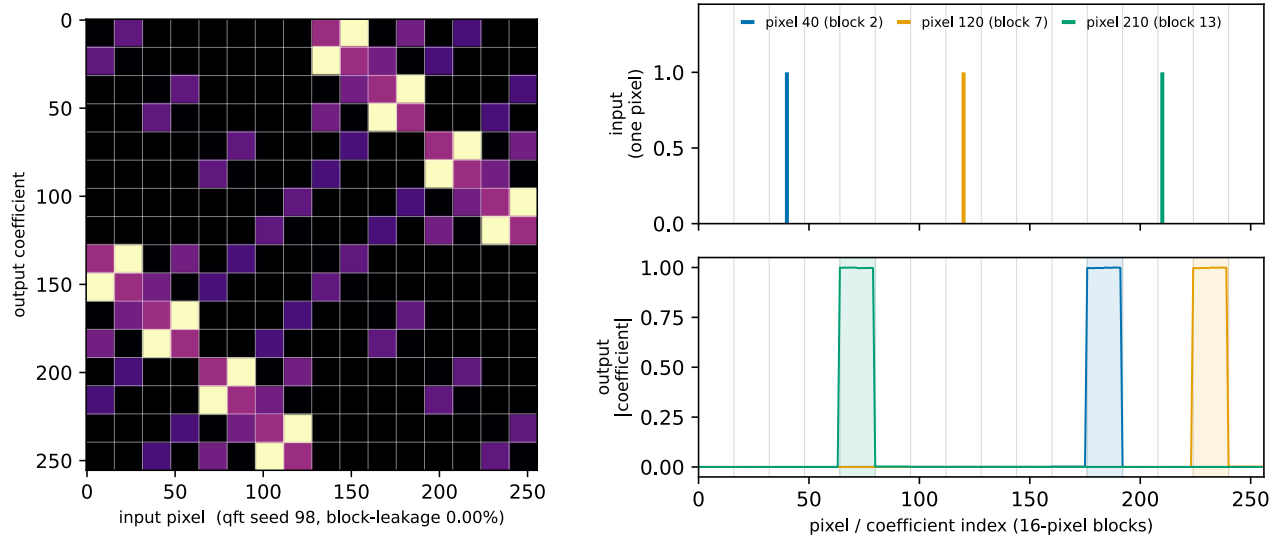


Figure 5: **Left:**  $\log|W|$  of the trained QFT’s 1-D operator factor, block-diagonal at the emergent 16-pixel block (off-block energy below 0.01%). **Right:** the same operator read one input at a time — a single input pixel (top) produces output (bottom) confined to a single 16-coefficient block ( $> 99.99\%$  of its energy), zero elsewhere. The output block is permuted from the input one by the QFT bit-reversal and the frozen-X gates; a global transform would instead spread each pixel across all 256 coefficients. The heatmap is these per-pixel responses stacked side by side.

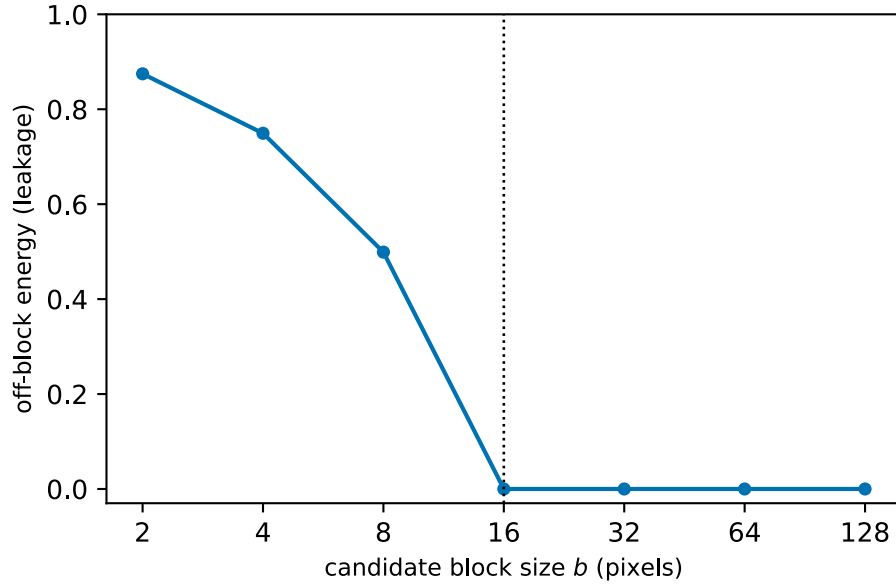


Figure 6: Off-block energy vs candidate block size — the knee at  $b = 16$  (dotted) shows the operator is block-structured at that specific scale and not below; the untrained closed-form QFT (not shown) stays high at every scale.

The same factorization is visible directly in frequency space.

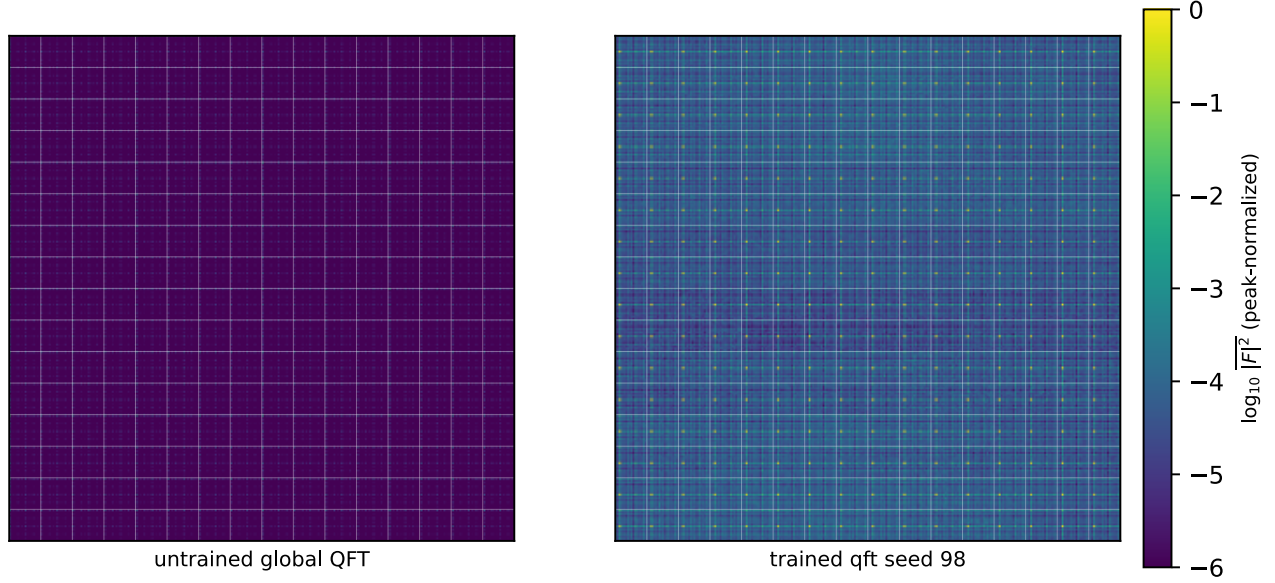


Figure 7: Mean test-set power spectrum (peak-normalized  $\log_{10} \overline{|F|^2}$ , averaged over the 50 DIV2K test images) of the untrained **global QFT** (left) and the **trained QFT** (right). The global transform packs energy into a single low-frequency lobe; the trained transform tiles into a  $16 \times 16$  grid of repeated sub-spectra — the block-periodic signature of a block transform, the same structure the  $8 \times 8$  block references produce in Fig. 4 of the main text.